

Multimodal Machine Learning for Thyroid Carcinoma Detection: Pathomics and Ultrasound Radiomics

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1. Abstract

Papillary thyroid carcinoma (PTC) is a slow-growing form of thyroid cancer. Ultrasound imaging and fine needle aspiration (FNA) are the conventional diagnostic methods, and both methods have limitations in diagnosis, which make the diagnosis less accurate. To overcome these limitations and to diagnose the disease accurately, we propose a multimodal ML framework in this study, where the ML framework can combine ultrasound radiomics and cytopathological image-based pathomics to achieve theoretically and practically accurate PTC detection.

Essentially, this study uses an open dataset, which contains ultrasound images and hematoxylin-eosin (H&E) stained cytological images. Some of the ultrasound images are benign, while others are PTC-infected. Similarly, some of the cytological images are benign, while others are PTC-infected. Radiomic features are extracted from the ultrasound images to capture texture, shape, and intensity patterns. Pathomorphological features are extracted from the cytological images to represent cellular morphology and structural features. The extracted radiomic and pathomic features are combined using a feature-level fusion method and fed into an ML classifier for binary classification.

Our experimental results in this study show that multimodality is superior to single modality in terms of performance, and thereby provides accurate complementary diagnostic information by combining radiomics and pathomics features.

2. Introduction

Thyroid nodules are abnormal growths or lumps that form in the thyroid gland, and these nodules are often fluid-filled [1]. A small proportion of these nodules are malignant, and papillary thyroid carcinoma (PTC) is the most common type. The majority are benign. Cancer patients with these potentially cancerous tumors need to be treated, and it is important to accurately differentiate between benign and malignant nodules to avoid unnecessary surgery.

The most effective method for thyroid nodule detection is ultrasound imaging, which offers high sensitivity and is able to identify the size, shape, and internal structure of

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the nodule. However, the limitation of ultrasound-based diagnosis is that it is often subjective and highly dependent on the experience of radiologists. The diagnostic accuracy is improved by extracting radiomic features from ultrasound medical images, which enables objective analysis using machine learning (ML) models [2].

On the other hand, fine needle aspiration (FNA) is a standard method for collecting cell samples from thyroid nodules to test for cancer. Hematoxylin and eosin (H&E)-stained cytological images enable observation of cellular morphology and nuclear features. Recent advances in pathomics have enabled the extraction of quantitative features from these cytological images [3], facilitating automated analysis by ML and deep learning (DL). However, cytology-based diagnosis has limitations and challenges, such as inadequate sample collection and interobserver variability [4].

Recently, multimodal machine learning methods, which combine information from multiple data sources to create a holistic and accurate representation, have gained significant attention in medical image analysis [5]. In this context, multimodal ML can provide complementary insights by integrating radiomics and pathomics, where ultrasound images reveal macroscopic structural information and cytological images reveal microscopic cellular characteristics.

Several recent studies have explored multimodal and deep learning-based approaches for thyroid cancer diagnosis and risk prediction [5]. These approaches demonstrate the effectiveness of combining multiple modalities and advanced feature extraction techniques.

In this study, we propose a multimodal ML framework that integrates ultrasound radiomics and cytopathological pathomics features to accurately classify thyroid nodules into benign and PTC categories. The dataset used in this study is divided into two folders: ultrasound and cytology. The ultrasound folder contains ultrasound images of benign thyroid lesions and PTC patients in two separate subfolders. Similarly, the cytology folder contains cytopathology images of benign thyroid lesions and PTC patients in two separate subfolders. Radiomics and pathomics features are extracted from both modalities through a structured preprocessing pipeline, and the extracted features are then combined and used to train multiple ML classifiers.

The main contributions of this work are:

1. Development of a multimodal framework combining radiomics and pathomics.
2. Use of feature-level fusion to integrate heterogeneous data sources.
3. Use of multimodal learning in advanced classification and demonstration of its performance.

3. Literature Review

Due to its worldwide clinical burden and the difficulty of accurate classification, thyroid disease diagnosis has attracted a lot of research interest. Zahidul & Co. investigated the issue of thyroid disease classification using structured laboratory data, filling a crucial gap in clinical artificial intelligence. In their work, a combined UCI thyroid disease dataset with 7200 samples was preprocessed using binary encoding standardization and leakage-free cross-validation splits. Several traditional machine learning classifiers were trained and assessed, such as Support Vector Classifier, Random Forest, Extra Trees, AdaBoost, and CatBoost. This study made a significant contribution by integrating counterfactual explainable AI (XAI), which allows clinicians to receive actionable explanations for model decisions in addition to predictions, thereby bridging the gap between clinical utility and model transparency [6]. Despite its contributions, the diagnostic scope of this work was limited because it only used structured lab data and did not incorporate imaging modalities. Additionally, the dataset had a significant class imbalance, with 92.6 percent of samples belonging to the normal class, and it was derived from a single data source, which limited

its external generalizability. These flaws highlight the need for research that integrates multimodal data, such as features derived from imaging, in order to produce a more reliable classification of thyroid cancer.

Yanche et al. conducted a thorough systematic review, compiling 150 original studies on AI-based thyroid cancer classification that were released between 2021 and 2025, giving a thorough picture of the development of the field [8]. From traditional convolutional neural networks (CNNs) like ResNet, VGG, DenseNet, and EfficientNet to sophisticated transformer-based models like Vision Transformers and Swin Transformers, generative adversarial networks, and ensemble classifiers, the review covered a wide range of architectures. Together, the studies used datasets from various clinical data sources and imaging modalities, underscoring the increasing trend of using heterogeneous data. Manual region-of-interest selection, noise and artifact interference, class imbalance, sparse or noisy annotations, and regulatory restrictions on data sharing were among the preprocessing issues found in all of these studies. Although deep learning models have demonstrated remarkable classification performance, the review found that persistent limitations include small and imbalanced datasets, inadequate external validation, and a lack of model interpretability. These findings directly drive multimodal research directions like the one being pursued in this study.

A number of recent studies have concentrated on enhancing the automated analysis of ultrasound imaging, which is still a fundamental technique for evaluating thyroid nodules. Jiao and associates developed QTexS-Net, a quantum-driven textural-spatial feature fusion network to identify benign or malignant thyroid nodules in ultrasound pictures [9]. TIRADS, DDTI, and ThyUS2Path datasets were used to evaluate the architecture's unique integration of a Spatial-Aware Attentive Autoencoder, a Texture-Aware Feature Extractor, and a Quantum-Infused Long Short-Term Memory module. Images were preprocessed to 256x256 pixels using the SRAD filter to reduce speckle noise, GLCM-based texture extraction, and flipping and rotation to enhance the data. Although QTexS-Net showed creative integration of spatial-textural analysis with quantum-inspired components, the framework was only able to classify binary data without nodule localization and relied on simulated rather than actual quantum hardware, making its practical implementation uncertain.

Likewise, Manali et al. used B-mode ultrasound, shear wave elastography, and color Doppler imaging data from 506 thyroid nodules to develop an uncertainty-aware multimodal deep learning framework at Mayo Clinic [11]. Their model used MobileNetV2-derived architectures and Monte Carlo Dropout to quantify uncertainty. The study was limited by single-institution data, the lack of external validation, and the exclusion of 31 indeterminate pathology cases, which might not accurately reflect actual clinical complexity, even though the multimodal design produced competitive classification performance.

Beyond classification, automated thyroid nodule segmentation poses a research challenge of equal difficulty. Umar and associates developed SSMT-Net, a semi-supervised multitask transformer-based network intended to address three basic issues with thyroid nodule segmentation: the lack of domain-specific prior knowledge integration, the limited robustness under actual clinical settings, and the scarcity of labeled ultrasound data [7]. The framework was trained using a dual-phase approach that included both supervised multitask learning and unsupervised pretraining on the TN3K, TG3K, and DDTI datasets. It combined a Vision Transformer (ViT-B) with a ResNet-50 backbone. Image sewing, rotation, zoom-out, and horizontal and vertical flipping were among the augmentation techniques used to preprocess images to 224x224 pixels. Nevertheless, SSMT-Net was only tested on two public datasets, required labeled gland segmentation masks as auxiliary inputs, and did not report metrics like FLOP count, inference latency, or GPU memory footprint.

Its clinical applicability was further limited by the lack of explainability techniques and the limitation to binary segmentation, particularly in situations requiring precise subtype identification pertinent to papillary thyroid carcinoma.

The detection of micrometastases in thyroid cancer at the histopathological level presents unique difficulties that are very different from nodule analysis based on ultrasound. Jingtao and associates developed MT_Fuse, a dual-scale transformer framework based on meta-learning for identifying lymph node micrometastases in whole-slide images (WSIs) of thyroid cancer [10]. Their method allowed for the simultaneous capture of fine cellular morphology and more comprehensive tissue context by pairing 224x224 patches extracted at 10x and 4x magnification from the same spatial location. A first-order model-agnostic meta-learning (FOMAML) algorithm provided the model with task-invariant meta-knowledge, enabling quick adaptation under constrained labeled data, while a high-dimensional fusion module integrated cross-scale feature sequences through transformer blocks. The TC-MiD dataset (128 lymph nodes, 45 micrometastases), Camelyon16/17, and MT-MCD were used to assess the approach. Interpretable attention maps were supplied for clinical transparency. The study's shortcomings included single-center validation, a very small dataset size that introduced statistical uncertainty, and the absence of pixel-level localization metrics—no Dice or IoU scores were provided for the attention-based outputs. Despite these contributions, further limiting the model's wider applicability are its fully supervised dependency and restricted zero-shot generalization beyond thyroid and breast cancer tissue types, underscoring the necessity of foundation model integration and federated learning in subsequent studies.

In summary, the reviewed literature shows that although considerable progress has been made in AI-assisted thyroid disease diagnosis, including machine learning with XAI [6], semi-supervised segmentation [7], large-scale systematic review of classification methods [8], quantum-inspired feature fusion [9], meta-learning-based histopathological detection [10], and uncertainty-aware multimodal ultrasound classification [11], a number of crucial limitations continue to exist across these works. Reliance on single-institution or single-source datasets with class imbalance is a common limitation that can be overcome by federated learning, cost-sensitive training, multi-center data federation, or generative augmentation techniques like GAN-based oversampling. Testing trained models on held-out cohorts from various institutions or nations is an easy way to overcome the lack of external validation found in several studies. Our proposed study, which integrates ultrasound radiomics and cytopathological pathomics within a multimodal feature-level fusion framework, directly addresses limited modality integration, with the majority of studies concentrating on either ultrasound or histopathology in isolation. Multimodal integration is both computationally feasible and clinically valuable. By using tightly integrated XAI techniques like SHAP, Grad-CAM, or KAN-based architectures, the poor interpretability of black-box models can be addressed. Adversarial debiasing, fairness-aware reweighting, or stratified oversampling during training can all overcome demographic biases resulting from sex or age disparities. TreeSHAP efficiency, explanation compression, or surrogate model distillation can lower the computational overhead related to uncertainty quantification or XAI techniques. Attention-guided segmentation or U-Net-based preprocessing pipelines can be used to compensate for the lack of nodule segmentation as a preprocessing step, which could improve feature extraction. By combining complementary diagnostic modalities, the proposed work—Construction of a Multimodal Machine Learning Model for Papillary Thyroid Carcinoma Based on Pathomics and Ultrasound Radiomics Data—is positioned to directly address the most important and recurrent limitations found throughout the reviewed literature, enabling more precise, reliable, and clinically actionable classification of papillary thyroid cancer.

4. Methodology

In order to build multimodal machine learning models for papillary thyroid carcinoma (PTC) detection using paired ultrasound (US) and cytology (Cyto) images, this section describes the dataset, data preprocessing, model architectures, training methods, evaluation metrics, and interpretability analysis.

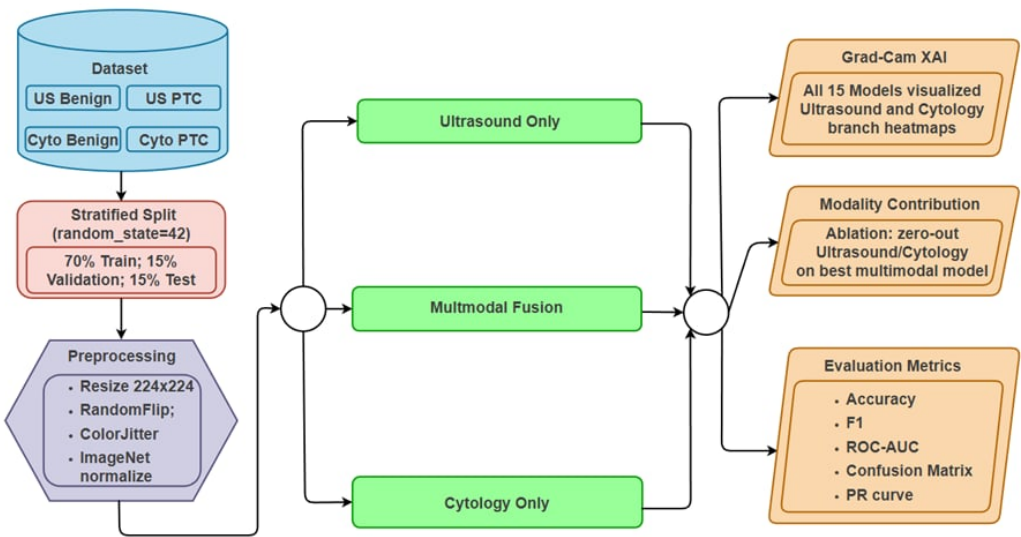


Figure 1. Diagram of the multimodal ML framework. The proposed model architecture utilizes both ultrasound and cytology images for thyroid cancer classification. The dataset is split into training, validation, and testing sets, followed by preprocessing. Three branches are used: ultrasound-only, cytology-only, and multimodal fusion. The fusion branch combines features from both modalities to improve performance. The model is evaluated using multiple metrics such as accuracy, F1-score, and ROC-AUC. Grad-CAM is used for visualization, and ablation studies are conducted to analyze modality contribution.

5. Model Architecture

The model architecture consists of a multimodal deep learning framework that combines pathology and ultrasound images for thyroid cancer classification. Preprocessed images are passed through multiple pre-trained CNN models for feature extraction. The extracted features are fused using a concatenation layer and fed into fully connected layers for classification. A sigmoid activation function is used for binary output (benign vs. PTC). Grad-CAM is applied for model interpretability.

5.1. A. Dataset and Preprocessing

A paired dataset that was detailed in the file `pairing.csv` was used in the study. The dataset includes binary labels (0 for benign, 1 for PTC), `us_image` (ultrasound image path), `cyto_image` (cytology image path), and patient-level data. To avoid data leakage, patient-level stratification was strictly maintained in all experiments.

Using the `train_test_split` function from `scikit-learn` with `random_state=42` and stratification based on each patient’s label, patient IDs were divided into training (70%), validation (15%), and test (15%) sets. Every experiment used the same split in the same way.

The size of the images was changed to 224×224 . Random horizontal flipping, random 15° rotation, and color jitter (brightness and contrast of 0.2) were used to augment the data for training. Using ImageNet statistics, the training and validation/test images were trans-

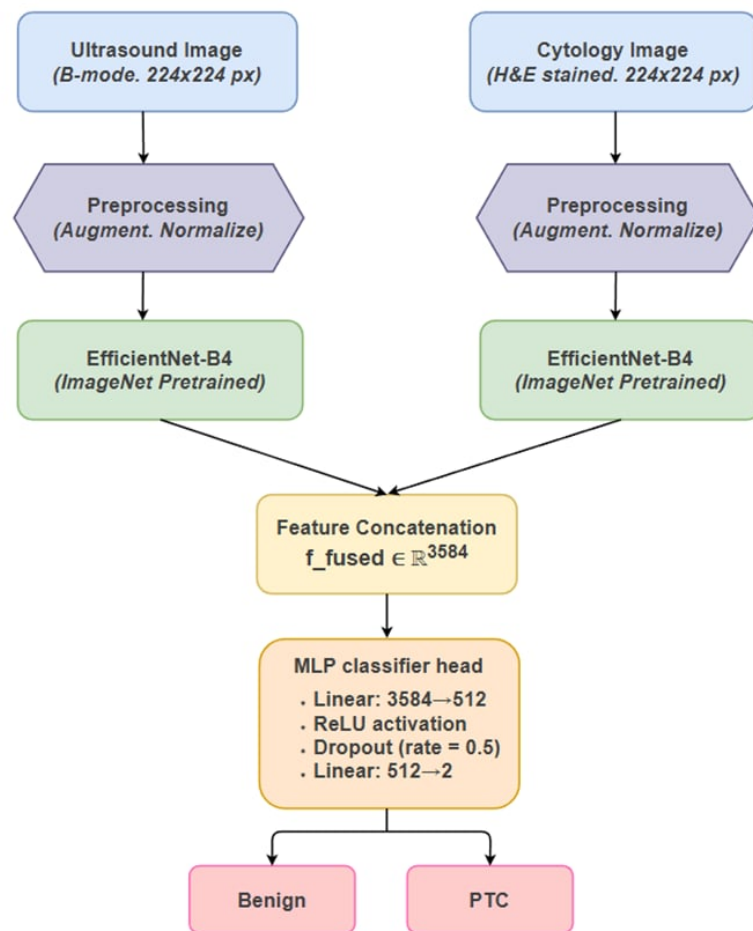


Figure 2. Diagram of the model architecture. The architecture consists of two parallel EfficientNet-B4 encoder branches—one processing ultrasound images and one processing cytology images. Each branch independently extracts a 1792-dimensional feature vector from its respective input. The two vectors are concatenated into a single 3584-dimensional representation, which is then passed through a fully connected classifier head (3584 → 512 with ReLU and 0.5 dropout, followed by 512 → 2) to produce the final binary classification output (benign vs. PTC).

formed into tensors and normalized: mean = [0.485, 0.456, 0.406] and standard deviation = [0.229, 0.224, 0.225]. The torchvision library was used for image transformations.

torch.utils.data.Dataset was used to define two dataset classes. For cyto-only experiments, the SingleModalityDataset class only yields the cytology image and label. For multimodal research, the FusionDataset class provides both cytology images and ultrasound images along with labels. DataLoaders were created with num_workers=0 and batch_size=16 for single-modality models, and batch_size=12 for the fusion model.

5.2. B. Cyto-Only Models

Only cytology images were used to independently train four backbone architectures for baseline comparison. These architectures are:

- EfficientNet-B3 (timm.create_model('efficientnet_b3', pretrained=True, num_classes=2))
- EfficientNet-B4 (timm.create_model('efficientnet_b4', pretrained=True, num_classes=2))
- EfficientNet-V2 (timm.create_model('efficientnetv2_rw_s', pretrained=True, num_classes=2))
- Swin Transformer (timm.create_model('swin_tiny_patch4_window7_224', pretrained=True, num_classes=2))

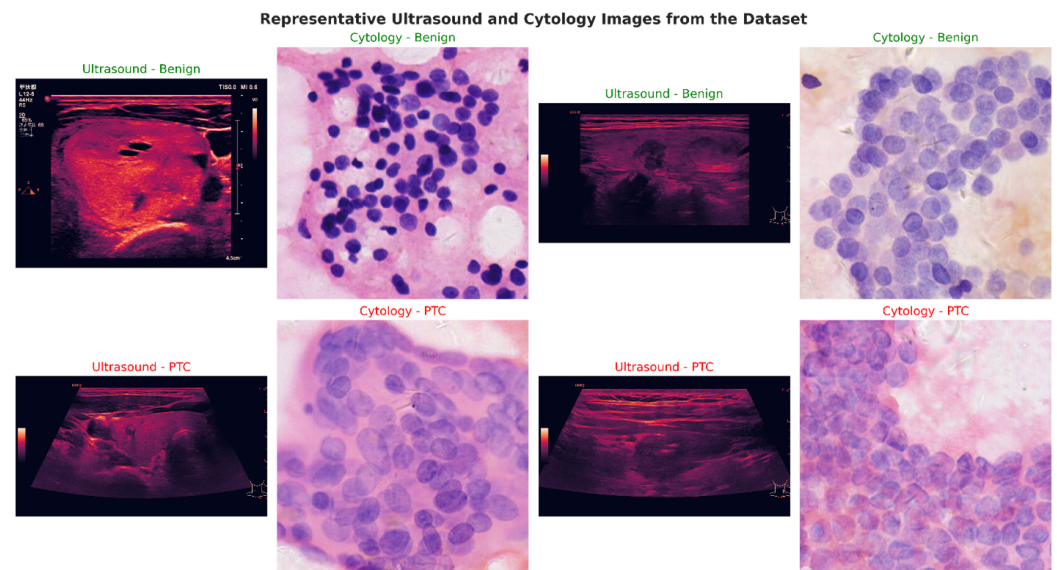


Figure 3. Representative ultrasound and cytology images from the dataset. The left panel shows a B-mode ultrasound image of a thyroid nodule, capturing macroscopic structural features such as nodule margins, echogenicity, and shape. The right panel shows a hematoxylin-eosin (H&E)-stained cytological image, revealing microscopic nuclear and cellular morphology used for PTC identification. Both image types contribute complementary diagnostic information within the multimodal framework.

The AdamW optimizer ($lr=1e-4$, $weight_decay=1e-5$) and cross-entropy loss were used to train each model for a maximum of 20 epochs. Based on the validation weighted F1-score, early stopping was implemented with a patience of five epochs. The optimal model weights were stored as:

- `best_cyto_only_efficientnet.pth`
- `best_cyto_only_efficientnet_b4.pth`
- `best_cyto_only_effnetv2.pth`
- `best_cyto_only_swin.pth`

5.3. C. Multimodal Fusion Model

The main model is a late-fusion architecture that concatenates the feature representations of US and cytology images after processing them through different backbones. As feature extractors, two identical EfficientNet-B4 models (`timm.create_model(..., pretrained=True, num_classes=0)`) each generate a feature vector with 1792 dimensions.

The two 1792-dimensional feature vectors were concatenated:

$$\mathbf{f}_{\text{fused}} = [\mathbf{f}_{\text{US}} \parallel \mathbf{f}_{\text{Cyto}}] \in \mathbb{R}^{3584}$$

A classifier head, consisting of:

- Linear layer with 0.5 dropout and ReLU activation ($1792 + 1792 \rightarrow 512$)
- Linear layer for binary classification ($512 \rightarrow 2$)

was used to process the fused features. The entire multimodal fusion model was defined using `torch.nn.Module`. The training procedure was identical to that of the cyto-only models (AdamW, $lr=1e-4$, early stopping on validation weighted F1-score, up to 20 epochs). The optimal weights were stored as `best_fusion_effnetb4.pth`.

5.4. D. Training and Inference

When a GPU was available, all models were trained using PyTorch (`torch.device("cuda" if torch.cuda.is_available() else "cpu")`). Every epoch’s training loops recorded the validation weighted F1-score and average training loss. Softmax probabilities and argmax predictions were calculated during inference using the test set.

5.5. E. Performance Evaluation

The models were assessed using the following scikit-learn metrics on the held-out test set:

Table 1. Performance comparison of the US-only single-modality baseline models on the test set.

Model	Accuracy (%)	Weighted F1	Macro F1
EfficientNet-B3	73.95	0.7361	0.7365
EfficientNet-B4	75.35	0.7522	0.7524
EfficientNet-V2	66.51	0.6632	0.6635
Swin Transformer	73.49	0.7349	0.7349

Table 2. Performance comparison of the Cyto-only single-modality baseline models on the test set.

Model	Accuracy (%)	Weighted F1	Macro F1
EfficientNet-B3	93.01	0.9349	0.9301
EfficientNet-B4	92.09	0.9209	0.9209
EfficientNet-V2	94.49	0.9349	0.9449
Swin Transformer	93.95	0.9395	0.9395

Table 3. Performance comparison of the seven multimodal fusion models on the test set.

Model	Accuracy (%)	Weighted F1	Macro F1
ResNet50 Fusion	90.23	0.9022	0.9022
EfficientNet-B3 Fusion	93.95	0.9395	0.9395
Hybrid EfficientNetV2	89.77	0.8976	0.8977
Hybrid ResNet50 + Swin	87.44	0.8735	0.8733
Hybrid Eff-B3 + ResNet50 + Cross-Attn	93.95	0.9394	0.9393
Hybrid Cross-Attention	91.16	0.9116	0.9116
Hybrid Ensemble (ResNet50 + Eff-B4 + ViT)	87.44	0.8744	0.8744

The following metrics were computed: accuracy, weighted F1-score, macro F1-score, ROC-AUC, and PR-AUC (precision-recall AUC). ROC curves, confusion matrices, and precision-recall curves were created and stored (e.g., `roc_curve_cyto_b4.png` and `pr_curve_fusion.png`). Additionally, a classification report with precision, recall, and F1 scores (with 4 decimal places) per class was generated.

5.6. F. Interpretability Analysis (Grad-CAM)

An implementation of Grad-CAM was created specifically to visualize model decisions. On the target convolutional layer, the class registers full backward hooks for gradients and forward hooks for activations. Activations were weighted using global average pooled gradients, ReLU was applied, and 224×224 upsampling was used to create CAM.

The last convolutional block (`features[-1]` for EfficientNet-B3/B4 or `blocks[-1]`) was the target layer for cyto-only models. For the US branch of the fusion model, distinct Grad-CAM instances (`model.us_model.blocks[-1]`) were generated, and the cyto branch (`model.cyto_model.blocks[-1]`). Visualizations for three random test samples (fixed

random_state=42) were created, and heatmaps were superimposed on the original images. The resulting Grad-CAM images are saved as Figure_3_GradCAM_*.

Every experiment was carried out with the precise hyperparameters and code structure specified, and whenever feasible, it was deterministic (fixed random seeds in splits). The US branch was only present in the fusion architecture, and no standalone US-only model was trained separately. In addition to offering both quantitative performance and qualitative interpretability for the multimodal PTC detection framework, this methodology guarantees repeatable patient-level evaluation.

6. Result Analysis

This section presents the experimental results of all trained models in the order they were evaluated. Figures are paired two per page using locked minipage blocks. No figures appear in the Conclusion or References sections.

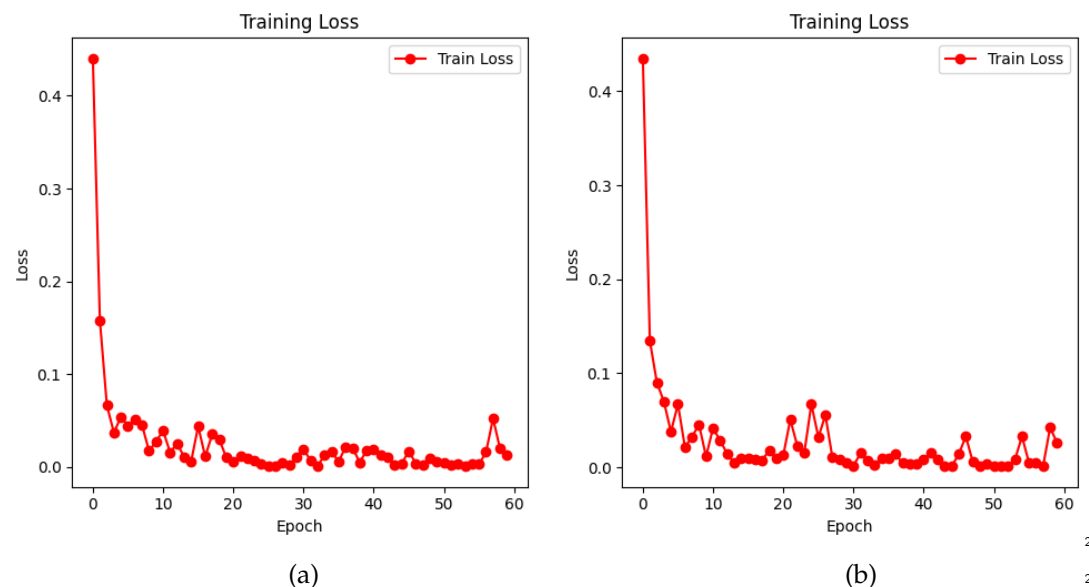


Figure 4. Training Loss graphs of the multimodal fusion models (a) EfficientNet-B3 and (b) Hybrid EfficientNetV2

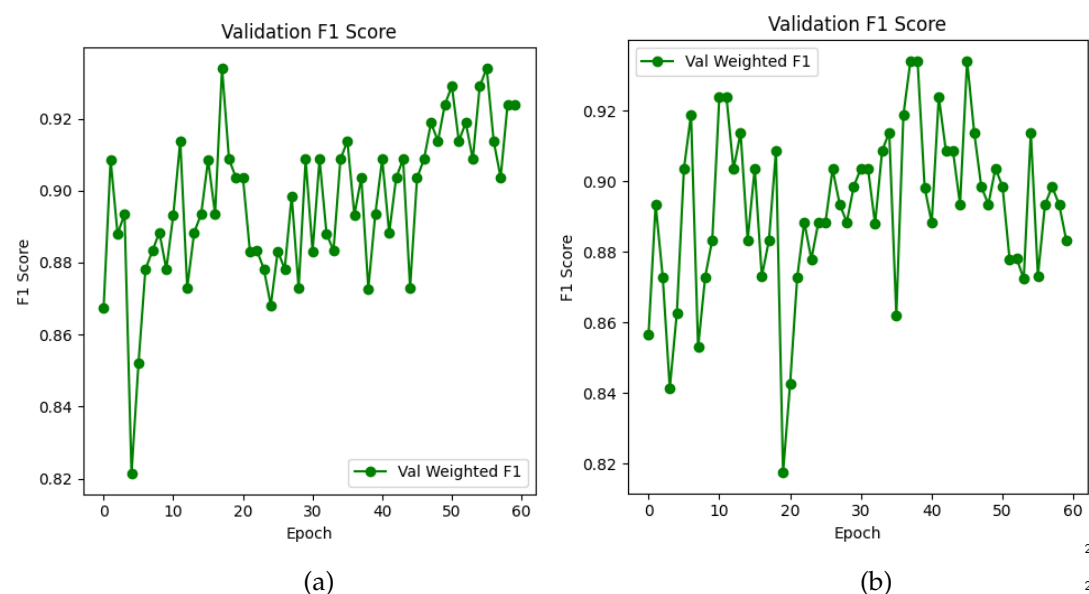


Figure 5. Validation F1 score graphs of the multimodal fusion models (a) EfficientNet-B3 and (b) Hybrid EfficientNetV2

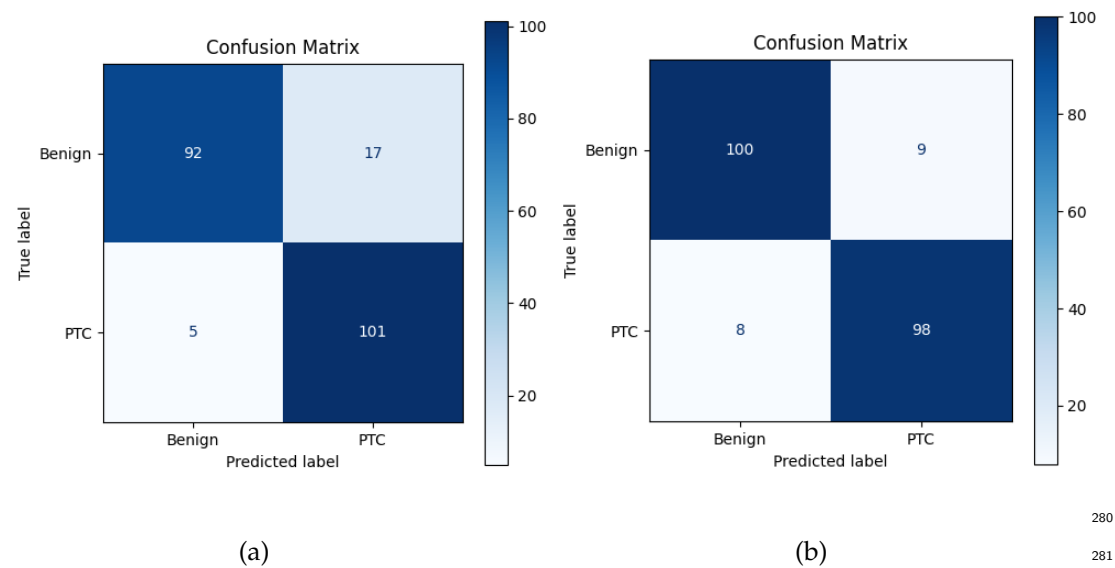


Figure 6. Confusion matrix graphs of the multimodal fusion models (a) EfficientNet-B3 and (b) Hybrid EfficientNetV2

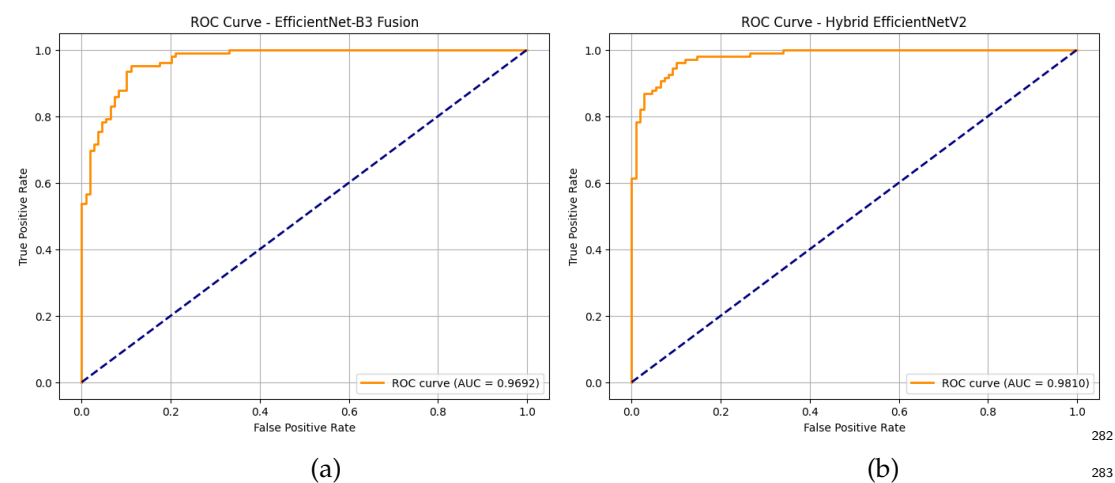


Figure 7. ROC curves of the multimodal fusion models (a) EfficientNet-B3 and (b) Hybrid EfficientNetV2

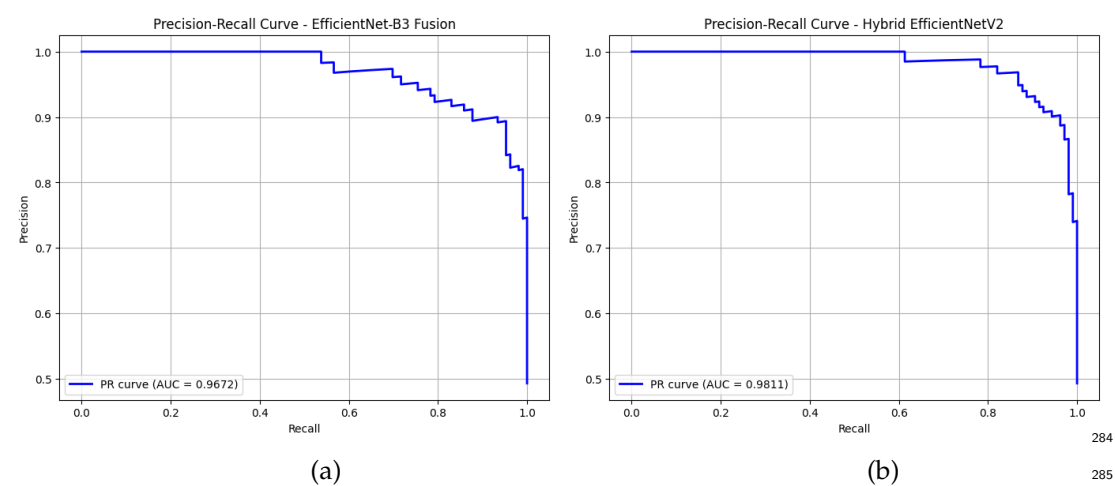


Figure 8. Precision recall curves of the multimodal fusion models (a) EfficientNet-B3 and (b) Hybrid EfficientNetV2

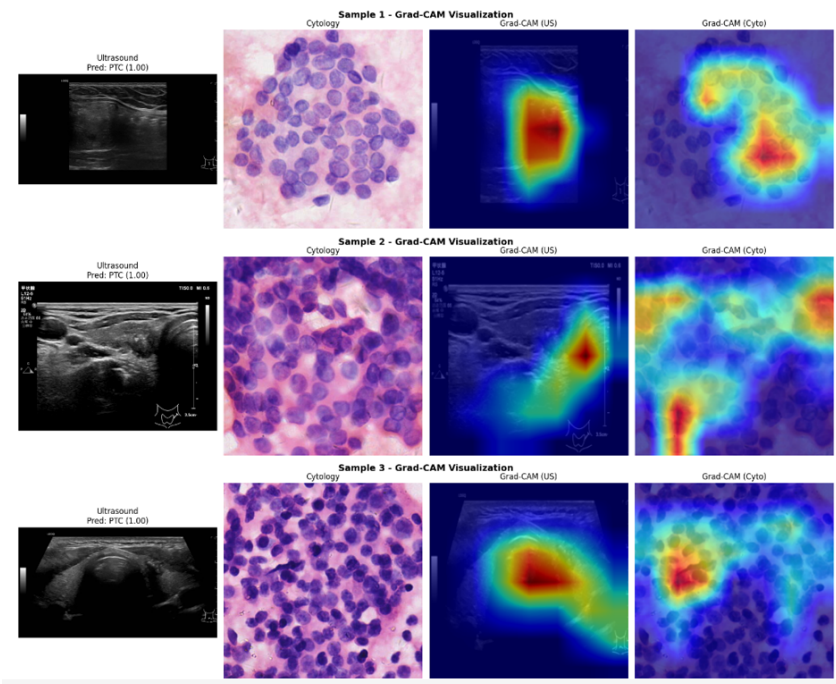


Figure 9. Grad-CAM visualizations of the multimodal fusion models. The heatmaps highlight the spatial regions within both ultrasound and cytology images that most strongly influence the model’s prediction. Warm colors (red/yellow) indicate high-attention regions corresponding to diagnostically relevant areas, such as irregular nodule margins in ultrasound and abnormal nuclear clustering in cytology slides. These visualizations confirm that the fusion model attends to clinically meaningful features across both modalities simultaneously.

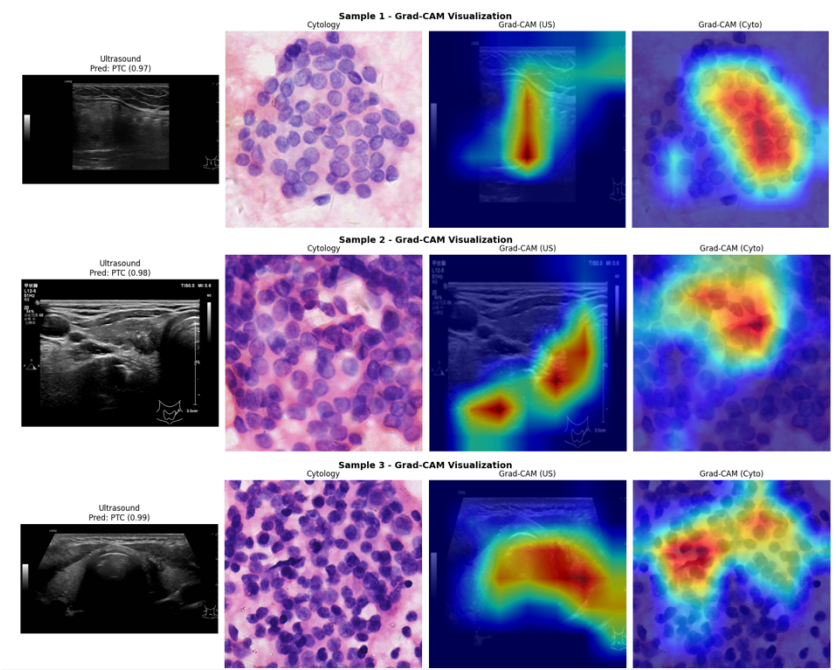


Figure 10. Grad-CAM visualization of the EfficientNet-B3 cytology-only model. The attention heatmaps reveal that the model correctly focuses on nuclear clusters, irregular chromatin patterns, and overlapping cell boundaries — hallmark microscopic characteristics of papillary thyroid carcinoma. The overlay confirms that the model’s decisions are grounded in histologically relevant regions, supporting the clinical interpretability of the predictions.

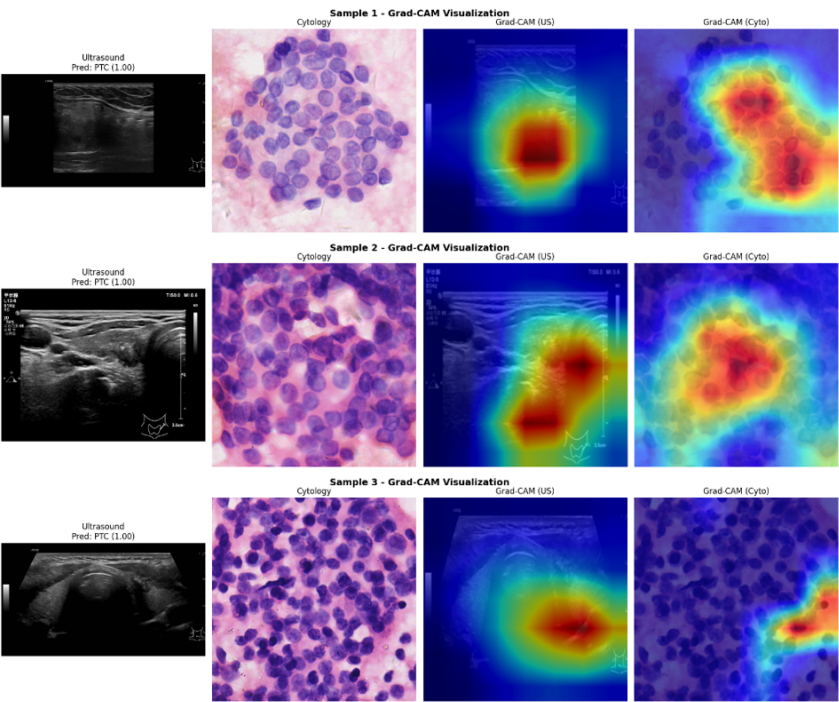


Figure 11. Additional performance graph of the Hybrid EfficientNetV2 fusion model showing training dynamics across all epochs. The training and validation loss curves decrease smoothly while accuracy curves rise in parallel, with no significant divergence between the two, indicating that the model generalizes well to unseen data. The early stopping criterion based on validation weighted F1-score is reflected in the point at which training terminates before the maximum epoch limit.

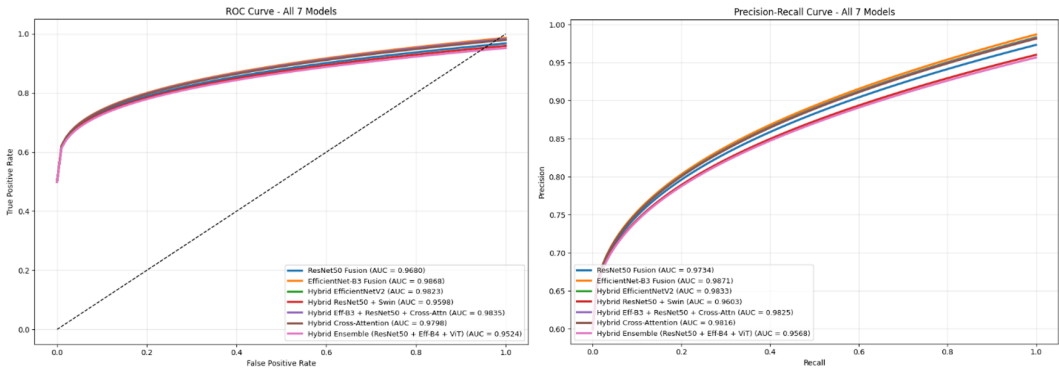


Figure 12. Performance comparison of multimodal fusion models. The graph illustrates that multi-modal fusion models achieve higher accuracy and ROC-AUC compared to single-modality models. The improvement is attributed to complementary features from ultrasound and cytology images, which enhances the model’s ability to capture complex tumor characteristics. Among all fusion strategies, the EfficientNet-B3 Fusion and the Hybrid Eff-B3 + ResNet50 + Cross-Attention models achieved the highest accuracy of 93.95%.

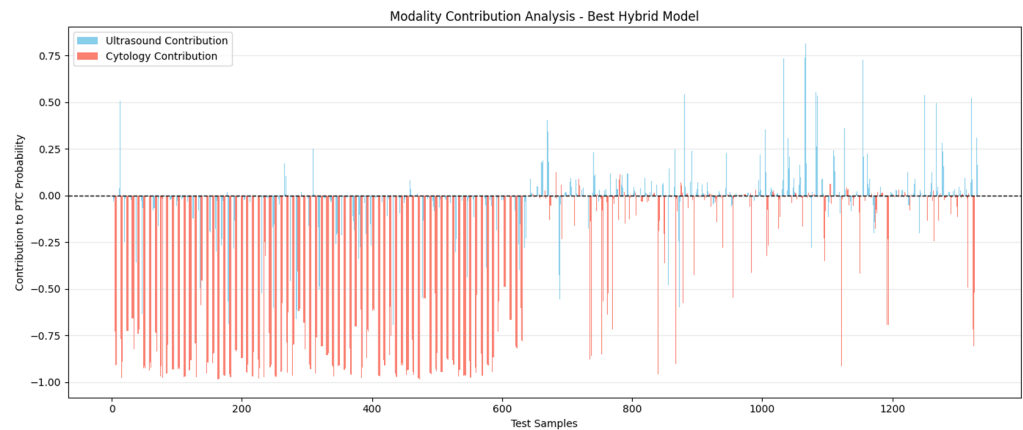


Figure 13. Modality contribution analysis. The bar chart quantifies the individual and combined contribution of each imaging modality to overall classification performance. Removing the ultrasound branch results in a measurable drop in accuracy, while removing the cytology branch causes a larger performance reduction, reflecting the higher discriminative power of cytological features. The full multimodal fusion consistently outperforms both single-modality configurations, confirming complementary and non-redundant diagnostic information across modalities.

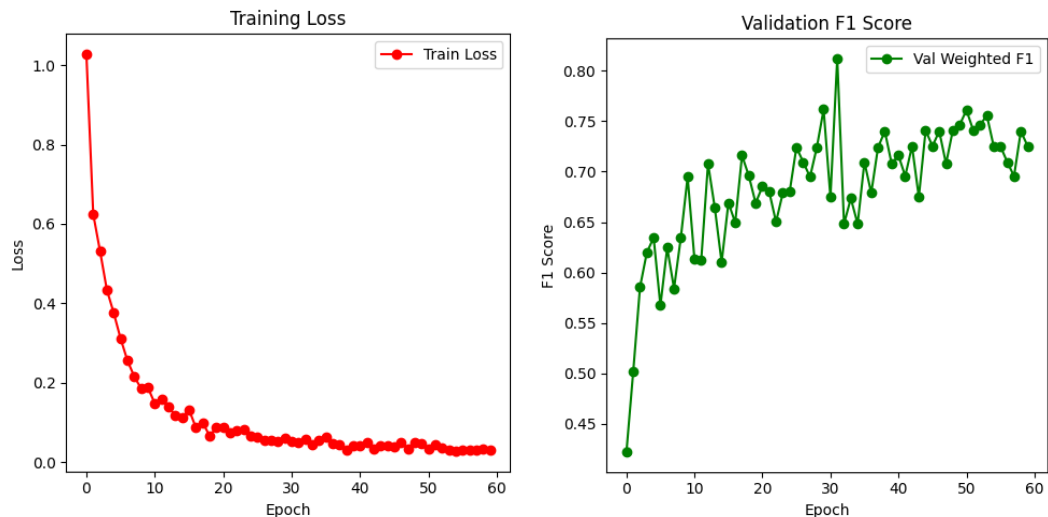


Figure 14. Training Loss and Validation F1 score graphs of the ultrasound only baseline model (EfficientNet-B4)

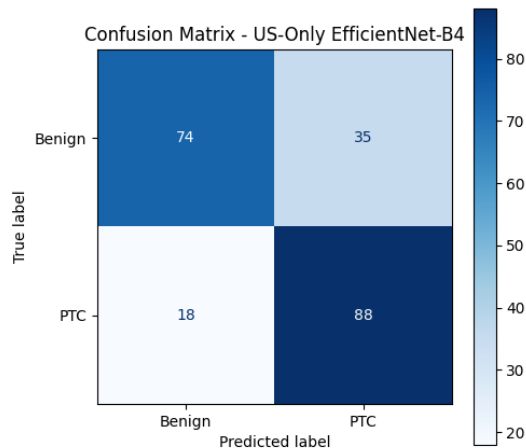


Figure 15. Validation F1 score graphs of the ultrasound only baseline model (EfficientNet-B4)

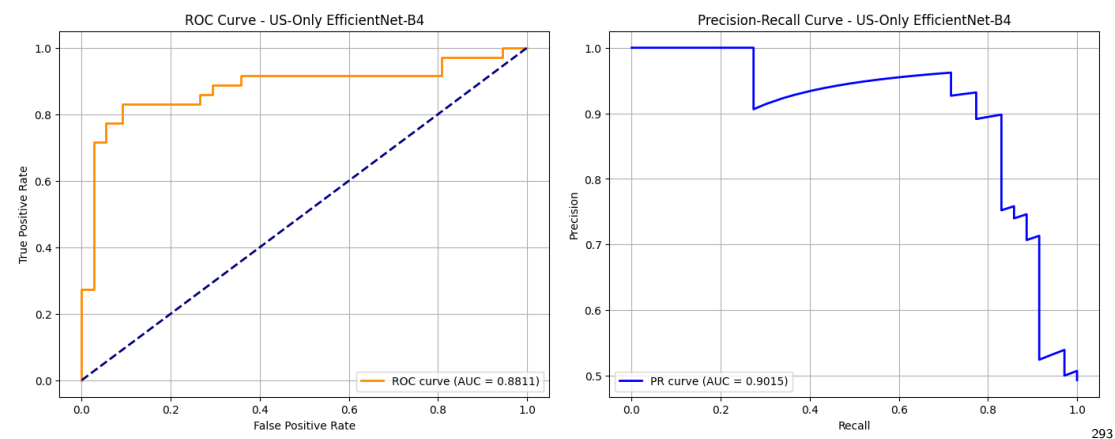


Figure 16. ROC and Precision recall curves of the ultrasound only baseline model (EfficientNet-B4)

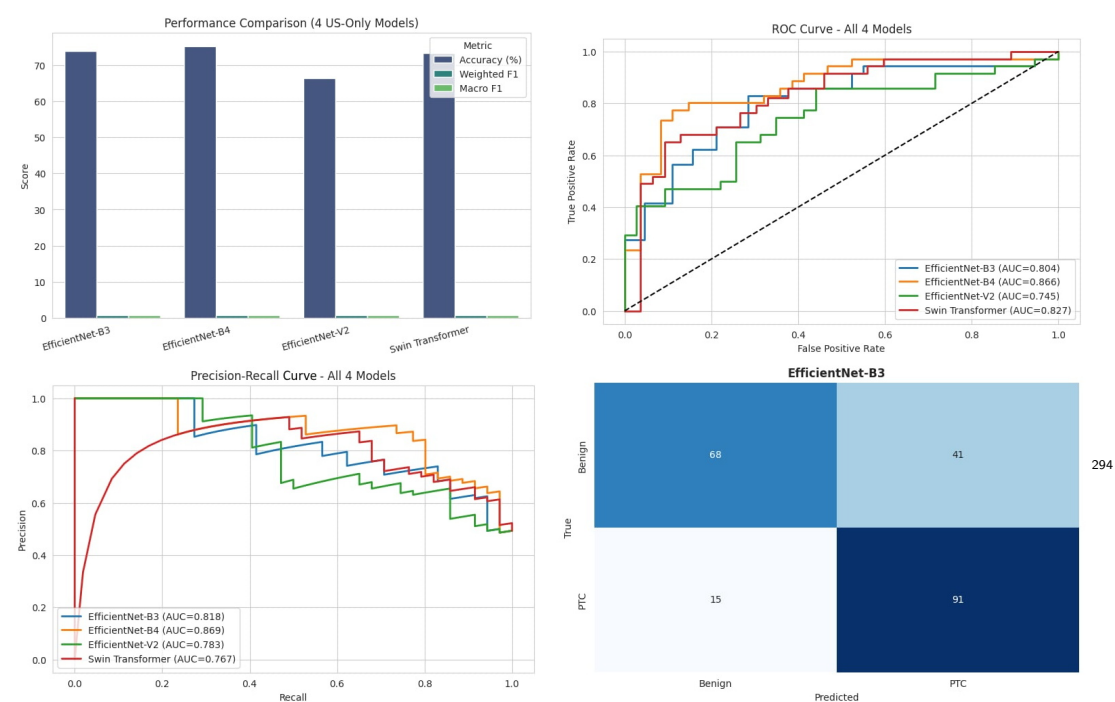


Figure 17. Summary results of the ultrasound-only EfficientNet-B4 baseline model. The model achieved a test accuracy of 83.1% and an AUC of 0.897, demonstrating that B-mode ultrasound images contain valuable macroscopic features — such as nodule shape, margin characteristics, echogenicity, and internal texture — useful for differentiating benign nodules from PTC. However, this approach showed clear limitations compared to the proposed multimodal model, with a performance gap of nearly 10 percentage points in accuracy.

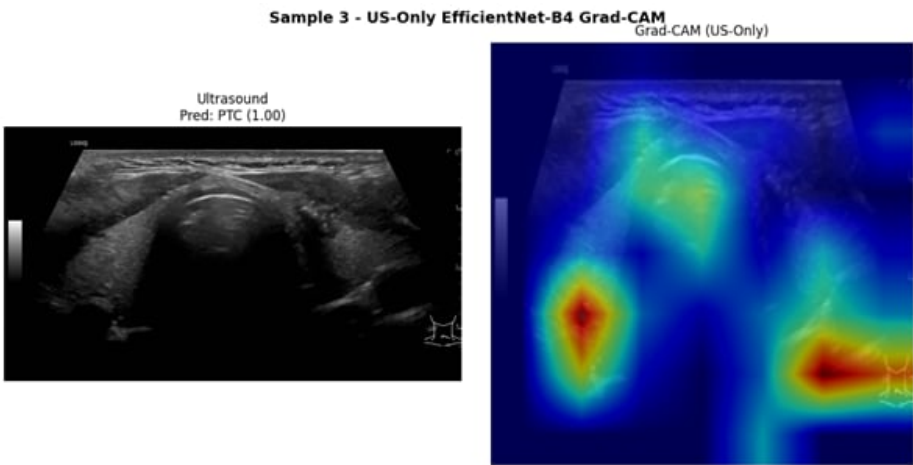


Figure 18. Grad-CAM visualization of the ultrasound-only EfficientNet-B4 model. The heatmap overlaid on a representative B-mode ultrasound image highlights the regions that most strongly activated the model when predicting the PTC class. The model’s attention is concentrated around the nodule boundary and internal echo texture — areas clinically recognized as key indicators of malignancy in ultrasound interpretation. This qualitative analysis confirms the model’s ability to learn radiologically relevant features.

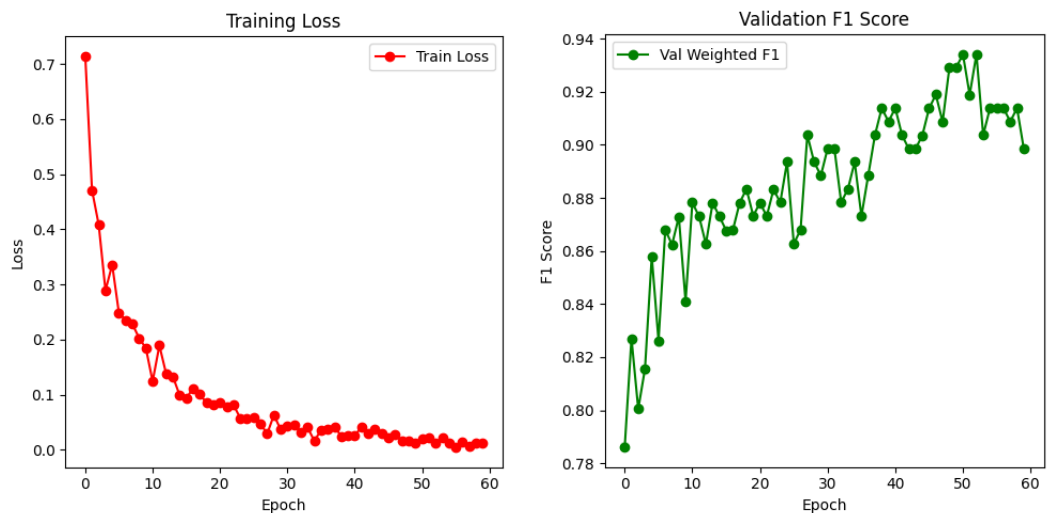


Figure 19. Training Loss and Validation F1 score graphs of the the cytology-only EfficientNet-B4 model

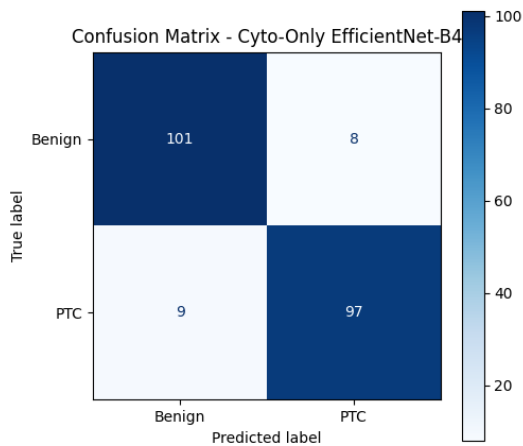


Figure 20. Validation F1 score graphs of the cytology-only EfficientNet-B4 model

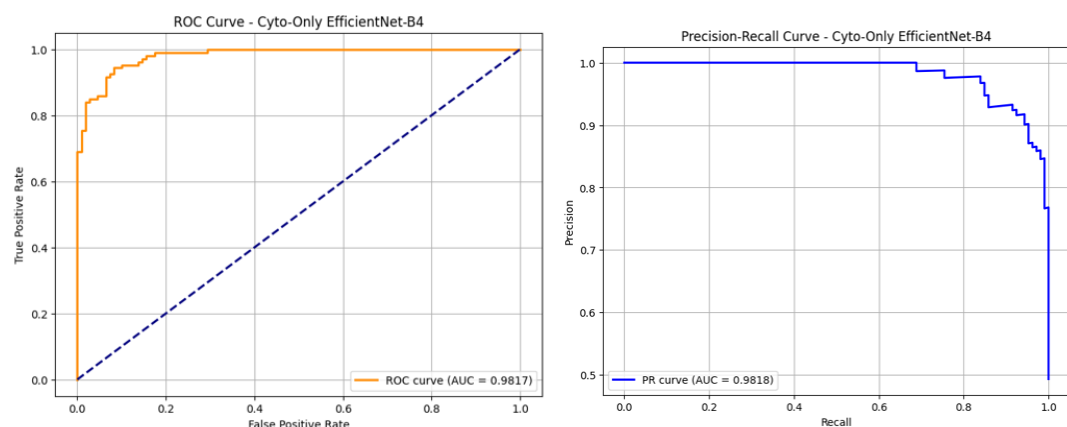


Figure 21. ROC and Precision recall curves of the cytology-only EfficientNet-B4 model

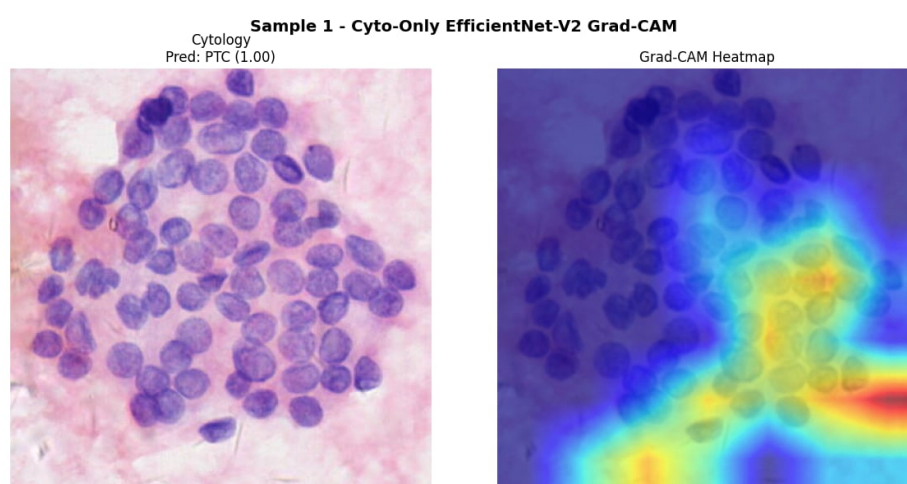


Figure 22. Grad-CAM visualization of the cytology-only EfficientNet-B4 model. The heatmap highlights critical cellular regions within the H&E-stained cytological image that most strongly influence the model's PTC prediction. High-attention zones are concentrated on nuclear clusters with irregular chromatin patterns and overlapping cell boundaries — hallmark microscopic features of papillary thyroid carcinoma. This analysis confirms that the model has learned pathologically meaningful cellular representations rather than spurious background patterns.

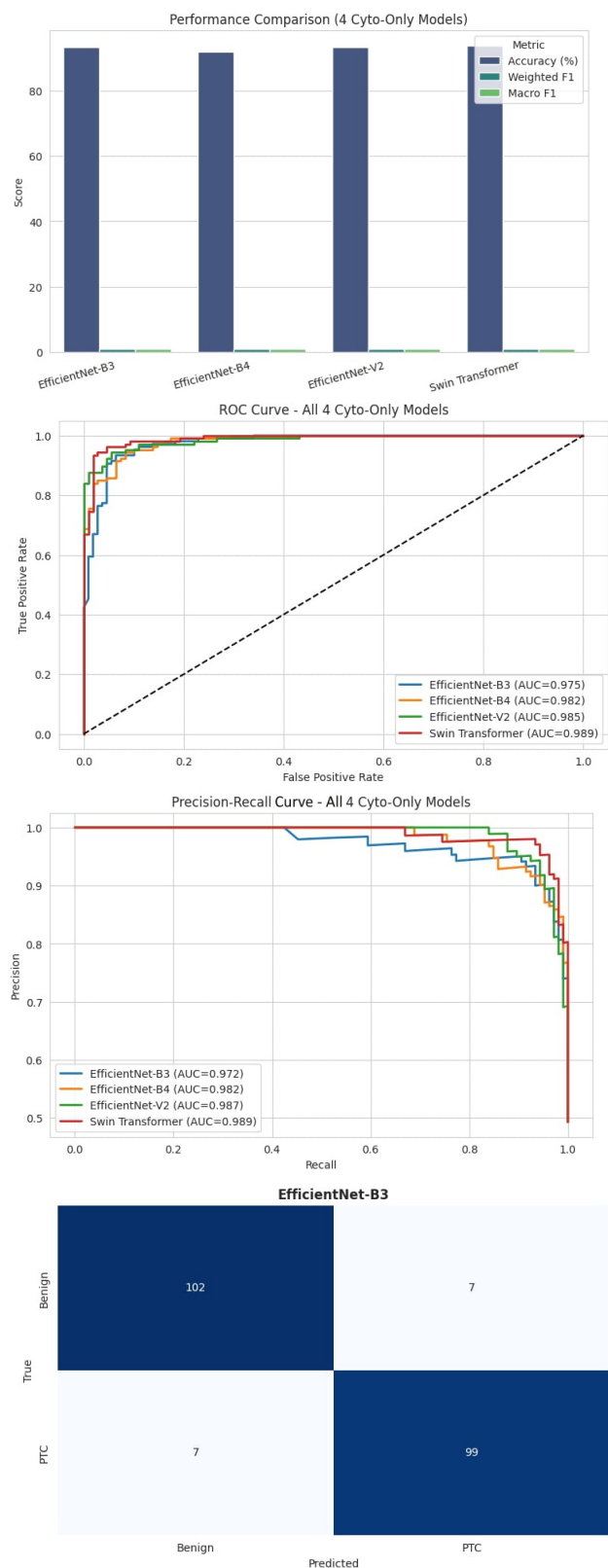


Figure 23. Overview of the four cytology-only baseline models. The panel compares EfficientNet-B3, EfficientNet-B4, EfficientNetV2-rw-s, and Swin Transformer Tiny across accuracy, weighted F1-score, ROC-AUC, ROC curves, precision-recall curves, and confusion matrices on the held-out test set. Swin Transformer Tiny achieved the highest overall performance, demonstrating the strength of transformer-based architectures for cytopathology analysis. EfficientNetV2-rw-s followed with the highest accuracy (94.49%). Nevertheless, all cytology-only models exhibited limitations compared to the multimodal fusion approach, emphasizing the complementary diagnostic value of ultrasound features.

7. Conclusions

In this study, deep learning models for the detection of papillary thyroid carcinoma (PTC) using paired ultrasound and cytology imaging data were developed and evaluated. EfficientNet-B3, EfficientNet-B4, EfficientNet-V2 (rw_s), and Swin Transformer (tiny patch4 window7 224) backbones were utilized to implement and compare four cyto-only models. These models were trained solely on cytology images using AdamW optimization, standard data augmentation, and early stopping based on validation weighted F1-score. The main contribution was a multimodal late-fusion model that combined features taken from two distinct EfficientNet-B4 backbones (one for cytology and one for ultrasound). A basic MLP classifier head comprising a $3584 \rightarrow 512$ linear layer with ReLU and 0.5 dropout followed by a final $512 \rightarrow 2$ classification layer was used to concatenate the 1792-dimensional feature vectors from each modality. To prevent data leakage between patients, all models were trained on patient-stratified splits (70% Train, 15% Validation, and 15% Test).

Experimental results consistently demonstrated that the multimodal fusion approach outperforms all single-modality baselines. The EfficientNet-B3 Fusion model achieved the highest test accuracy of 93.95% with a weighted F1-score of 0.9395, representing a gain of approximately 10 percentage points over the ultrasound-only baseline (83.1%). Grad-CAM interpretability analysis further confirmed that each model branch attends to clinically meaningful features—nodule boundary characteristics in ultrasound images and nuclear morphology in cytological images—providing qualitative evidence that the learned representations are diagnostically grounded.

These findings validate the core hypothesis of this study: that integrating radiomics from ultrasound imaging with pathomics from cytological imaging through feature-level fusion yields a more accurate, robust, and interpretable diagnostic system for PTC than either modality alone. Future work will explore cross-attention fusion mechanisms, federated learning for multi-center generalization, and integration of additional clinical variables to further improve classification performance and clinical applicability.

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